

Annotation of Clarification Requests in the CoDraw Dataset

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1 Introduction

This project is about annotating instruction clarification requests in the CoDraw dataset. CoDraw is a game where a teller describes an illustration made of a set of cliparts to a drawer who does not see the image. During the game, the teller has one chance to peek at the drawer's image, but is otherwise unable to see the drawing of the other person. The cliparts that are used to complete the illustration can have one of two orientations (facing left or right) and one of three sizes (small, normal, large). The goal of this game is to reconstruct the original image as closely as possible. To fulfill the task, it is often required that the drawer requests further information about the given instructions if the information was unclear, incomplete, or ambiguous. Such requests are called instruction clarification requests and this annotation project is focusing on them.

2 Setting

The annotation was performed by one person during three months for eight hours per week, using a graphical user interface. The annotator is 26 years old and female, speaks German as her native language and has a C1 level in English. She is a student of linguistics, took a course about dialogue modelling and she was provided with relevant literature about clarification requests specifically in order to prepare for this annotation task. She familiarised herself with the cliparts used in CoDraw and dialogues in CoDraw.

The potential clarification requests by the drawer are displayed together with the preceding and subsequent utterance by the teller. Some of the utterances do not have a context which can occur if it is the first or last turn in the dialogue or if the utterance is empty or occurs in different contexts. For the first round, another annotator already selected for each of the 13700 utterances by the

drawer whether it is an instruction clarification request. The second annotator now has to agree or disagree with this decision. If it is decided that the utterance is not a clarification request, no further steps are required for this case. Otherwise, it is necessary to answer multiple questions. To better explain the annotation guideline, example sentences similar to the clarification requests in CoDraw will be added.

First, the mood of the clarification request is selected. It can be *declarative* (*I'm not sure where to place the dog*), a *polar question* (*Is it small?*), an *alternative question* (*Is the cat facing right or left?*), a *wh-question* (*Which type of tree?*), *imperative* (*Tell me the size of the girl*), or *other*. Multiple moods can be chosen, because one utterance can include multiple clarification requests, as in: *Which direction is the girl facing? Left or right? Is she small?*

Next, it is annotated whether the clarification request refers to the preceding utterance or not and whether the drawer gets a response by the teller in the subsequent utterance.

After that, the question *Is it possible to know which clipart the CR refer to?* is asked and can be answered with *unknown*, *many*, *two*, or *one*. *Many* is chosen if more than two cliparts are mentioned in the utterance and at least one of them is identifiable, while *unknown* means that it is not possible to know which objects the utterance is referring to and the number of cliparts is unknown as well.

Unless *unknown* was chosen, the cliparts then need to be selected in the order they occurred. Up to five objects can be specified. For example, this includes different types of balls, trees, glasses, animals, clothing, food, toys, objects in the sky or the ground, and a girl and a boy. *Background* can

also be selected as object, if the drawer refers to the sky or the ground, which is the only visible image for the drawer at the beginning of the game. *Ambiguous* can be chosen, if it is not entirely clear which clipart is being referred to, for example, if *boy* or *girl* is described with *child*.

Lastly, it is annotated what the clarification request is about, choosing one or more out of six categories.

- Information may be requested about *size*, *position*, or *direction* (orientation) of a clipart.
- *Object disambiguation* is selected if the drawer does not know which clipart the teller is referring to. Clarification requests indicating this category are, for example: *Which tree?/Cat or dog?/The purple or black glasses?*
- Another category is person disambiguation. This category is used, when it is not clear to the drawer whether the other person is talking about the girl or the boy. However, most of the time, *person disambiguation* is selected, when the drawer requests information about the type of girl or boy. There are five different expressions for each (angry, sad, happy, happy and smiling with teeth showing, and shocked) and seven different positions. The latter are composed of arm and leg positions.
 - The children can be sitting cross-legged or with legs to the side, standing, jumping, running, or kicking with one leg.
 - The arm positions are slightly different for boy and girl, but in general they have either one or both arms up, to the side or pointing downwards.

Typical instruction clarification requests are *Is she sad or happy?/Both arms up?/Is he angry and kicking?/Is the girl standing or sitting?*

- The last relation is called *relation to other*. It is selected when two or more cliparts are compared with each other. *Is her head as big as the sun?/Is the bear left of the tree?/Does the cat face the boy?* and similar questions belong to this category. As mentioned before, more than one category can be selected. For example, for *Is the boy as small as the girl?*, both *size* and *relation to other* would be

selected.

3 Decisions

In the course of annotating, some further decisions were made, in order to better capture information about the instruction clarification requests and to clarify cases that were not evident.

Identifying Clarification Requests

Usually, utterances are only considered clarification requests, if the object, that the drawer is requesting information about, was already mentioned before. Therefore, we decided not to annotate *Is he wearing a hat?*, since *hat* was most likely not written about before and the drawer is showing self-initiative. However, the decision is not always clear. If the teller says: *She is kicking to the left* and the drawer asks: *Is she kicking a ball?*, then this seems more likely to be a clarification request. The drawer might be incorrectly interpreting what is meant by *kicking*, which is only referring to the position of the girl and not necessarily related to kicking an object.

Moods

- Originally, it was intended to annotate only one clarification request per utterance. However, we noticed that one utterance commonly consisted of two or more related or unrelated clarification requests. For this reason, we decided to allow the annotation of multiple moods per round. A typical utterance of the drawer could be similar to *Is the bear close to the tree? What size? Is he small or large?*
- When a sentence provided context to a clarification request, it was also annotated as a clarification request. For example, the teller could say: *There is a dog next to the tree*. The drawer could answer with: *I don't see a dog. Is it a cat?* In this case, the first phrase would be annotated as declarative.
- Some questions did not fully fit the standard format of the defined moods, as they were written in informal language, potentially containing errors concerning grammar and spelling. Therefore, we established rules to classify the mood of these clarification requests.

- *Is the girl next to the bear? In front of the bear? (Is the girl next to or in front of the bear?)*

Looking at the two questions separately, both parts can be considered as polar questions. However, the teller would not be able to answer with *yes* or *no*, so this cannot be the correct mood. For cases like this, we decided to classify them as alternative questions, as the drawer is expecting the teller to choose between the two options.

- *The ball is small? (Is the ball small?)* and *Baseball? (Is it a baseball?)* and similar questions are treated as polar questions, as these questions can be answered with *yes* or *no*.

- *Are they sitting or? (Are they sitting?)*
We consider this type of question to be a polar questions because no alternative is provided.

Identifying Cliparts

- Originally, it was only possible to select the different clip arts or the background as the objects that the clarification request is about. Later on, we added groups (*tree_group*, *cloud_group*, *glasses_group*, *ball_group*, *hat_group*) to solve the problem of ambiguity of the descriptions. A drawer could ask, for example: *Which hat?* In this case, it is not clear which hat they are referring to. Sometimes they are also mentioning an object that was described before: *Does the bear cover the tree?*. We can see that the drawer already knows about the tree, however, the annotator would have to choose *tree_group* here as well, as they do not have further information about the object. We also added *ambiguous*, for objects that are not identifiable.
- Only cliparts to which the drawer refers are selected. They may ask *Is it a pine tree or an apple tree?*, so both objects need to be selected, even though the teller is talking about only one of those two cliparts.

Number of Cliparts

- *The dog is on the right side. - Are they all on the right side?* In this example, the drawer is asking about multiple cliparts, however, the

number is not known. We select *many* instead of *unknown* in this case, since one object (dog) is known and *all* requires a number greater than two. As cliparts, we would choose *dog* and *ambiguous* with the latter summarizing all other objects that the drawer is referring to.

- Someone might also ask *Which tree?* or *Is the tree medium?*, where we decided to select *tree_group*. While the drawer is talking about one specific tree, the group includes three trees. Since we focus on the drawer's utterance, we chose *one* as the correct number of cliparts for similar cases.
- *Unknown* and *many* can overlap, so we decided that the first mentioned category means that we do not know what the drawer is talking about and whether it is one or more objects. For *many*, the utterance needs to be referring to more than two objects and at least one clipart has to be known.

Clarification Request Categories

- If someone asks *Can you describe the boy to me?*, it could fit into multiple categories. The drawer could want to know about size, position, direction, or facial expression and pose. However, the categories are not explicitly stated, so we decided to always select person disambiguation (referring to facial expression and pose) because it seems most likely to be the requested information.
- In questions like *Her right side or my ride side?* the drawer wants to know what *right side* means, whether it is referring to their own perspective or the perspective of boy, girl, or an animal (girl in this case). We decided to select both *position* and *relation to other* as categories for similar questions.
- *Is the girl facing the boy?* and *Is the head as big as the sun?* also belong to two categories: *relation to other* and *directions/size*.
- Although all corresponding categories are supposed to be annotated, for sentences like *Is she behind the sandbox?* only *relation to other* is annotated due to an initial misinterpretation of the category *relation to other* by the annotator.

4 Errors

- Due to a limitation of a library, the order of the different clarification requests within one utterance could not be preserved.
- The images *balloon* (referring to the hot air balloon) and *air balloons* (referring to the bundle of balloons) were accidentally mixed up throughout the annotation. [Note from BM: this has been fixed programmatically in the OSF .tsv file.]